

ClearPath - A wearable obstacle detection device for the visually impaired that provides direct feedback.

Background - For the longest time, the visually impaired have relied on a walking cane or someone else at detecting obstacles in their path. This device would be able to be worn (possibly as glasses) by a user and it would have sensors that work similarly to how sensors work inside of a tesla car. They would play a little chime for the user to hear if there is an obstacle within a certain threshold along their path. This project aims to build “ClearPath”, a wearable pair of glasses that are fitted with many sensors that have been designed to communicate to the user any hazards and provide direct feedback to them.

Project Needs:

1. A design of what the glasses could look like. They have to be comfortable enough and fitted with a long lasting battery so they are secure enough for the user the entire day.
2. Find a way to develop detection level sensors that are able to not only locate obstacles, but also provide direct instantaneous feedback back to the user.
3. Possibly create an app to allow modification and customizations of settings for the user. Make sure it is user-friendly for the visually impaired community. On the other hand, find a way to incorporate bluetooth directly into the glasses.
4. Go through a vast amount of testing to ensure that they are not only safe, but also efficient. Testing can be done with the help of visually impaired volunteers or by simulating with a blindfold.

Technical Considerations:

1. Use an arduino or something similar for sensor integration
2. Find a way to source real time sensors small enough that can be used for the prototype
3. Implement code for feedback system
4. Build app for user calibration
5. Look into finding a database to use to store logs of users information

Deliverables:

1. Functional working prototype
2. Possible app for calibration of glasses
3. Documentation providing reasoning for all decisions made regarding sensors to ensure safety
4. Proof of testing to provide comfort for users and those who wish to learn about it.